

Discrete Choice

Binary Choice

Recall from logistic regression:

1. $y \in \{0, 1\}$ = Dependent variable (binary outcome, yes or no, qualitative)
2. p = Number of predictor variables (excluding the constant 1)
3. $x_1, \dots, x_p$ = Independent variables (predictors)

We are interested in estimating the linear model:

$$y^* = \beta_0 + \sum_{j=1}^p \beta_j x_j + \epsilon = \boldsymbol{\beta}' \mathbf{x} + \epsilon$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_p \end{bmatrix}.$$

However we do not observe y^* (which is a latent variable), but rather we observe a variable y where:

$$y = 1 \text{ if } y^* \geq 0, \text{ and } y = 0 \text{ otherwise.}$$

We can view y as a binary random variable which represents a random choice (choose or do not choose) where:

$$\begin{aligned} P(y = 1 | \mathbf{x}, \boldsymbol{\beta}) &= P(y^* \geq 0 | \mathbf{x}, \boldsymbol{\beta}) \\ &= P(\boldsymbol{\beta}' \mathbf{x} + \epsilon \geq 0 | \mathbf{x}, \boldsymbol{\beta}) \\ &= P(\epsilon \geq -\boldsymbol{\beta}' \mathbf{x} | \mathbf{x}, \boldsymbol{\beta}) \\ &= 1 - F(-\boldsymbol{\beta}' \mathbf{x}) \\ &= F(\boldsymbol{\beta}' \mathbf{x}) \text{ (if the distribution of } \epsilon \text{ is symmetric around 0)} \end{aligned}$$

where $F(\cdot)$ is the cumulative distribution function of the continuous random variable ϵ , namely $\epsilon \sim F(\cdot)$.

If we set $F(z) = \frac{e^z}{1+e^z}$ we get,

$$P(y = 1 | \mathbf{x}, \boldsymbol{\beta}) = \frac{e^{\boldsymbol{\beta}' \mathbf{x}}}{1 + e^{\boldsymbol{\beta}' \mathbf{x}}}$$

which is the binary logit model and identical to the logistic regression model. We could use other F , for example $F(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$ (standard normal) which results in a binary probit model.

Multinomial Choice

Problem setup:

1. K = Number of alternatives (available choices)
2. $y \in \{1, 2, \dots, K\}$ = Dependent variable (choice made)
3. $\mathbf{x}_1, \dots, \mathbf{x}_K$ = Independent variables (observable attributes of the alternatives encoded in vectors)

Here we assume the utility for alternative k is random:

$$U_k = \boldsymbol{\beta}' \mathbf{x}_k + \epsilon_k \text{ for } k = 1, \dots, K$$

where $\boldsymbol{\beta}$ is the weight vector associated with the observable attributes of the alternatives. Here ϵ_k is an additive noise term that models part of the random utility that is not captured by the linear term $\boldsymbol{\beta}' \mathbf{x}_k$. Again, we do not observe U_k (utility), but rather we observe a variable y (choice made) which can be defined as follows:

$$y = k \text{ if } U_k = \max_{l=1, \dots, K} U_l \text{ (utility of alternative } k \text{ is highest).}$$

Throughout, we will assume that there are ties with zero probability. Assume that ϵ_k are independent and identically distributed with Gumbel (type 1 extreme value) distribution given by:

$$F(z) = e^{-e^{-z}} \quad \forall z \in \mathbb{R}$$

and density function given by:

$$f(z) = e^{-z} e^{-e^{-z}} \quad \forall z \in \mathbb{R}.$$

It was shown by McFadden (1974) that,

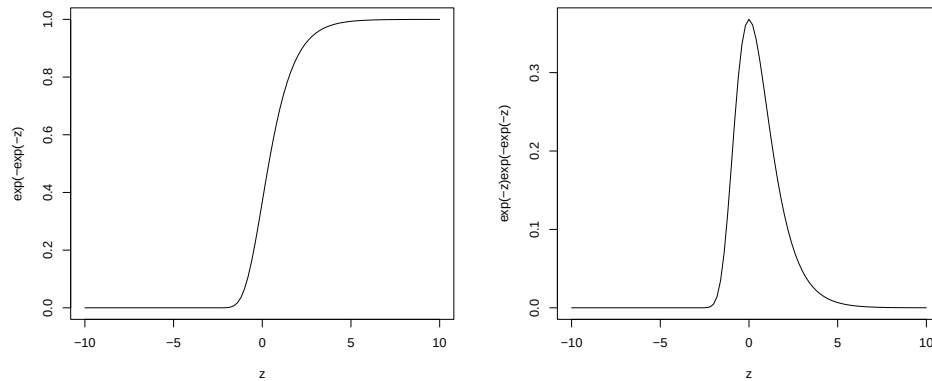


Figure 0.1: Cdf and pdf of Gumbel distribution

$$P(y = k | \mathbf{x}_1, \dots, \mathbf{x}_K, \boldsymbol{\beta}) = \frac{e^{\boldsymbol{\beta}' \mathbf{x}_k}}{\sum_{l=1}^K e^{\boldsymbol{\beta}' \mathbf{x}_l}}$$

This model is known as the conditional logit model (also referred to as the multinomial logit model).

Derivation of logit probability

Define $v_k = \beta' \mathbf{x}_k$ for all $k = 1, \dots, K$.

$$\begin{aligned}
P(y = k | \mathbf{x}_1, \dots, \mathbf{x}_K, \beta) &= P(v_k + \epsilon_k \geq v_l + \epsilon_l \ \forall l \neq k) \\
&= P(\epsilon_l \leq v_k - v_l + \epsilon_k \ \forall l \neq k) \\
&= \int_{-\infty}^{\infty} P(\epsilon_l \leq v_k - v_l + z \ \forall l \neq k) \underbrace{e^{-z} e^{-e^{-z}}}_{\text{Density of } \epsilon_k} dz \\
&= \int_{-\infty}^{\infty} \underbrace{\prod_{l:l \neq k} P(\epsilon_l \leq v_k - v_l + z)}_{\text{From independence of } \epsilon_l} e^{-z} e^{-e^{-z}} dz \\
&= \int_{-\infty}^{\infty} \prod_{l:l \neq k} e^{-e^{-(v_k - v_l + z)}} e^{-z} e^{-e^{-z}} dz \\
&= \int_{-\infty}^{\infty} e^{-z} \prod_{l=1}^K e^{-e^{-(v_k - v_l + z)}} dz \\
&= \int_{-\infty}^{\infty} e^{-z} e^{-\left(\sum_{l=1}^K e^{-(v_k - v_l + z)}\right)} dz \\
&= \int_{-\infty}^{\infty} e^{-z} e^{-e^{-z} \left(\sum_{l=1}^K e^{-(v_k - v_l)}\right)} dz \\
&= \int_{\infty}^0 -e^{-t} \sum_{l=1}^K e^{-(v_k - v_l)} dt \quad (\text{change of variables with } t = e^{-z}, dt = -e^{-z} dz) \\
&= \int_0^{\infty} e^{-t} \sum_{l=1}^K e^{-(v_k - v_l)} dt \\
&= \frac{1}{\sum_{l=1}^K e^{-(v_k - v_l)}} \\
&= \frac{e^{v_k}}{\sum_{l=1}^K e^{v_l}}
\end{aligned}$$

Independence of irrelevant alternatives: Multinomial Logit

Consider two alternatives k and l and compute the ratio of the choice probabilities:

$$\frac{P(y = k)}{P(y = l)} = \left(\frac{e^{\beta' \mathbf{x}_k}}{\sum_{t=1}^K e^{\beta' \mathbf{x}_t}} \right) / \left(\frac{e^{\beta' \mathbf{x}_l}}{\sum_{t=1}^K e^{\beta' \mathbf{x}_t}} \right) = e^{\beta' (\mathbf{x}_k - \mathbf{x}_l)}$$

Independence of irrelevant alternatives: The ratio of the choice probabilities of two alternatives is unaffected by the other alternatives in the choice set. Adding in a new alternative does not change the ratio of $P(y = k)/P(y = l)$. Another way to express this is that the new alternative gains shares proportionately from the choice shares of existing alternatives in the set.

Example: Blue bus/Red bus paradox.

Alternatives: Car, Red bus

Suppose a commuter chooses between a car and a red bus with equal probability. Then: $P(\text{Car}) = P(\text{Red bus}) = 0.5$

Assume a blue bus is added. It is reasonable to assume that commuters do not care about the color of the bus.

Alternatives: Car, Red bus, Blue bus

$P(\text{Car}) = 0.5, P(\text{Red bus}) = P(\text{Blue bus}) = 0.25$: Reasonable prediction

$P(\text{Car}) = P(\text{Red bus}) = P(\text{Blue bus}) = 0.33$: MNL prediction

This happens because blue bus and red bus are perfect substitutes which is not captured by the MNL model. On the other hand, the MNL model has other useful properties; for example as we will see next, estimating the parameters is efficiently possible.

Maximum Likelihood Estimation

Let n be the number of observations (individuals). We are interested in estimating the linear model:

$$U_{ik} = \beta' \mathbf{x}_{ik} + \epsilon_{ik} \text{ for } i = 1, \dots, n, k = 1, \dots, K$$

where U_{ik} is the random utility of individual i for alternative k and $\mathbf{x}_{ik}$ is the attribute vector of individual i and alternative k . This attribute vector could include attributes that depend only on the individual i (for example: income, age, family size), only on the alternative k (for example: price, volume) or on the individual and alternative pair (for example: price of a taxi ride from home to school for a student). Our dataset consists of $(y_{i1}, \dots, y_{iK}, \mathbf{x}_{i1}, \dots, \mathbf{x}_{iK})$ for $i = 1, \dots, n$ where $y_{ik} \in \{0, 1\}$ with $\sum_k y_{ik} = 1$ for each $i = 1, \dots, n$. Here we code the choices as follows:

$$y_{ik} = \begin{cases} 1 & \text{if individual } i \text{ chooses } k \\ 0 & \text{otherwise.} \end{cases}$$

The choice probability is given by:

$$\underbrace{P(y_i = k)}_{\text{Probability that individual } i \text{ chooses } k} = P(U_{ik} = \max_{l=1, \dots, K} U_{il})$$

In some cases, we will include alternative specific constants ASC_k for each alternative by modeling the utility as follows:

$$U_{ik} = ASC_k + \beta' \mathbf{x}_{ik} + \epsilon_{ik}.$$

The problem of estimating β (coefficients) that maximizes the likelihood of the observations is given as:

$$\max_{\beta} \underbrace{L(\beta)}_{\text{Likelihood given } \beta} = \max_{\beta} \prod_{i=1}^n \prod_{k=1}^K P(y_i = k | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iK}, \beta)^{y_{ik}}$$

Taking logarithms, this problem can be reformulated as maximizing the log-likelihood.

$$\max_{\beta} \underbrace{LL(\beta)}_{\text{Log-likelihood given } \beta} = \max_{\beta} \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log(P(y_i = k | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iK}, \beta)) = \max_{\beta} \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log \left(\frac{e^{\beta' \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}}} \right)$$

The objective function is concave and this optimization problem can be efficiently solved using Newton's method. This is one of the few models for discrete choice where the maximum likelihood problem is efficiently solvable.

Concavity of log-likelihood

Let us start by showing the concavity of a single term in the log-likelihood function:

$$\underbrace{\log \left(\frac{e^{\beta' \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}}} \right)}_{\text{to show concavity}} = \underbrace{\beta' \mathbf{x}_{ik}}_{\text{Linear in } \beta} - \underbrace{\log \left(\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}} \right)}_{\text{show convexity of this function}}$$

The key function we need to analyze is the **log-sum-exp** function $h : \mathbb{R}^K \rightarrow \mathbb{R}$ which is defined by:

$$h(\mathbf{w}) = \log \left(\sum_{l=1}^K e^{w_l} \right).$$

The gradient vector and the Hessian matrix are computed as:

$$\nabla h(\mathbf{w}) = \frac{1}{S(\mathbf{w})} s(\mathbf{w}) \text{ and } \nabla^2 h(\mathbf{w}) = \frac{1}{S(\mathbf{w})^2} \left(S(\mathbf{w}) \underbrace{\text{diag}(s(\mathbf{w}))}_{\text{diagonal matrix}} - s(\mathbf{w}) s(\mathbf{w})' \right)$$

where

$$s(\mathbf{w}) = \begin{bmatrix} e^{w_1} \\ \vdots \\ e^{w_K} \end{bmatrix} \text{ and } S(\mathbf{w}) = \sum_{l=1}^K s_l(\mathbf{w}) \text{ where } s_i(\mathbf{w}) = e^{w_i}$$

Let us verify that the Hessian matrix is positive semidefinite, namely $\mathbf{v}' \nabla^2 h(\mathbf{w}) \mathbf{v} \geq 0$ for all $\mathbf{v}$. This can be shown as follows:

$$\begin{aligned} \mathbf{v}' \nabla^2 h(\mathbf{w}) \mathbf{v} &= \frac{1}{S(\mathbf{w})^2} (S(\mathbf{w}) \mathbf{v}' \text{diag}(s(\mathbf{w})) \mathbf{v} - \mathbf{v}' s(\mathbf{w}) s(\mathbf{w})' \mathbf{v}) \\ &= \frac{1}{S(\mathbf{w})^2} \left(\left(\sum_{l=1}^K s_l(\mathbf{w}) \right) \left(\sum_{l=1}^K v_l^2 s_l(\mathbf{w}) \right) - \left(\sum_{i=1}^K v_i s_i(\mathbf{w}) \right)^2 \right) \\ &\geq 0 \quad (\text{from Cauchy-Schwarz inequality which states } (\sum_i a_i b_i)^2 \leq \sum_i a_i^2 \sum_i b_i^2). \end{aligned}$$

Since the Hessian matrix is positive semidefinite, the log-sum-exp function is convex. The convexity of $\log \left(\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}} \right)$ then follows from the observation that it is the composition of h (the log-sum-exp function which is convex) and an affine function of β . Lastly the concavity of $LL(\beta)$ follows since it is defined by the nonnegative sum of concave functions which must be concave.

Quality of fit

1. AIC (Akaike Information Criterion)

$$AIC = -2LL(\hat{\beta}) + 2p$$

where $\hat{\beta}$ is the estimated coefficient vector (this might include alternate specific constants), $LL(\hat{\beta})$ is the log-likelihood at the estimated coefficients and p is the number of parameters that are estimated. Observe that if we add the utilities of all alternatives by a single constant, the choice probabilities are not changed. Smaller the AIC, the better.

2. Likelihood ratio index (McFadden's index)

$$\rho = 1 - \frac{LL(\hat{\beta})}{LL(0)}$$

where $\hat{\beta}$ is the estimated coefficient vector and $LL(0)$ refers to the log-likelihood when all the parameters are set to 0 (no model; except possibly alternative specific constants if included). The likelihood ratio index ranges from 0 (estimated model is no better than having parameters set to 0 except possibly alternative specific constants) to 1 (estimated model perfectly predicts the choices observed).

3. Percent correctly predicted

This identifies the alternative with the highest probability for each individual observation and determining whether or not this was what the actual choice made (this is a very coarse measure of accuracy).

Special case (Multiclass Classification/Multinomial Logistic Regression): Assume we want to categorize data (for example images encoded as vectors) into one of several categories. This gives rise to a multinomial logistic regression or multiclass classification problem where:

$$P(\mathbf{x} \text{ is assigned to class } k) = \frac{e^{\beta'_k \mathbf{x}}}{\sum_{l=1}^K e^{\beta'_l \mathbf{x}}}$$

Here β_k is a vector of weights corresponding to category k .

Willingness to pay

In discrete choice modeling, where utilities of items are modeled with different attributes (including price), the effects of the variables may be gauged by a quantity called the willingness to pay. Suppose β_1 is the coefficient for attribute 1 and β_2 is the coefficient of attribute 2 (Price). Note that β_2 will be typically negative since more the price, lesser the utility. Say β_1 is positive. For a particular level of attribute 1, say, x_1 and a price x_2 , we have utility:

$$U = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \text{other terms.}$$

The question is that for a one unit increase in attribute 1, how much price are we willing to pay (wtp) so that the utility remains the same. Suppose this amount is Δ . Then:

$$U = \beta_0 + \beta_1(x_1 + 1) + \beta_2(x_2 + \Delta) + \text{other terms.}$$

Hence we obtain, the willingness to pay for unit increase in attribute 1 is

$$\text{wtp} = \Delta = -\frac{\beta_1}{\beta_2}.$$

Summary of output from mlogit in R

Package: mlogit

Function: mlogit (use dfix to create data for mlogit)

1. Frequencies of alternatives - This provides the frequency of choice in the data.
2. Coefficients - This provides estimates of coefficients, standard error of coefficients, t-value and p-value ($P > |t|$). To access these use `model$coefficients` or `coefficients(model)`. For example, you can access the standard error by `coefficients(summary(model))["Std. Error"]`.
3. Log-likelihood - Log-likelihood of fitted model

Mixed Logit

This model capture features that multinomial logit does not capture such as random taste variation.

Standard Logit: $U_{ik} = \beta' \mathbf{x}_{ik} + \epsilon_{ik}$

Mixed logit: $\tilde{\beta}$ is modeled as a random parameter vector with $U_{ik} = \tilde{\beta}' \mathbf{x}_{ik} + \epsilon_{ik}$

Standard Logit

$$P(y_i = k) = \frac{e^{\beta' \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}}}$$

Mixed Logit

$$P(y_i = k) = \int \frac{e^{\beta' \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta' \mathbf{x}_{il}}} f(\beta) d\beta$$

In the mixed logit model, $f(\beta)$ is the density function of $\tilde{\beta}$ which leads to the integral of logit probabilities. As a result the mixed logit model is computationally more challenging and one has to make use of simulation based optimization methods to estimate the parameters. The likelihood function is no longer concave in this setting and finding a global maximum is not as easy. From an estimation perspective, the goal is to find the parameters θ that define the density function $f(\beta|\theta)$ where the functional form $f(\cdot)$ is given but parameters θ are unknown.

The simulation-optimization method works (roughly) as follows:

1. Make an initial guess about the parameter θ .
2. Draw R samples from the distribution $f(|\theta)$; call them β_r for $r = 1, \dots, R$ and compute for each sample

$$P_{ik}^r = \frac{e^{\beta_r' \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta_r' \mathbf{x}_{il}}}$$

3. Next approximate the choice probability by the sample average:

$$P(y_i = k) \approx \sum_{r=1}^R \frac{P_{ik}^r}{R}$$

4. Plug in this approximation into the log-likelihood objective function to find a simulated log-likelihood

$$\underbrace{SLL(\boldsymbol{\theta})}_{\text{Simulated log-likelihood given } \boldsymbol{\theta}} = \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log \left(\sum_{r=1}^R \frac{P_{ik}^r}{R} \right).$$

where $y_{ik} = 1$ if i th individual chose the k th item.

5. Now do an update step on the parameters by moving in the direction of $\nabla SLL(\boldsymbol{\theta})$ and repeat the simulation and optimization update steps till a convergence criterion is met. It is also possible to use an approximation of the Hessian matrix in the update step (BFGS algorithm in nonlinear optimization).

For example, if we assume that $\tilde{\boldsymbol{\beta}}$ is normally distributed and independent across entries, then the parameter vector $\boldsymbol{\theta}$ consists of the means and standard deviations. In mixed logit models with repeated choices (panel data), where i : individual, k : alternative, t : observation (here every individual does a total of T choice tasks), the choice probability will be given by:

$$P(y_{i1} = k_1, \dots, y_{iT} = k_T) = \int \prod_{t=1}^T \left(\frac{e^{\boldsymbol{\beta}' \mathbf{x}_{ik_i t}}}{\sum_{l=1}^K e^{\boldsymbol{\beta}' \mathbf{x}_{il t}}} \right) f(\boldsymbol{\beta}) d\boldsymbol{\beta}$$

Panel data needs to account for the fact that the errors are correlated for the same individual over time.

Additional Notes

1. Softmax function:

The softmax function is often used in neural networks and LLMs. It is defined as:

$$\text{softmax}_{i,T}(\mathbf{v}) = \frac{e^{v_i/T}}{\sum_{l=1}^K e^{v_l/T}} \text{ for } i = 1, \dots, K.$$

Properties of the function include:

- All values are strictly positive
- The values sum up to one
- Converts a tuple of K real numbers into a probability distribution over K possible outcomes
- As (temperature) $T \uparrow \infty$, the probability distribution converges to an uniform distribution over the K outcomes.
- As (temperature) $T \downarrow 0$, the probability distribution converges to a distribution that is concentrated only on the outcome(s) with the largest v_i value(s). To see this, assume outcome i^* has the maximum value v_{i^*} . Namely $v_{i^*} > v_l$ for all $l \neq i^*$. Then:

$$\begin{aligned} \text{softmax}_{i^*,T}(\mathbf{v}) &= \frac{e^{v_{i^*}/T}}{e^{v_{i^*}/T} + \sum_{l \neq i^*} e^{v_l/T}} \\ &= \frac{1}{1 + \sum_{l \neq i^*} e^{(v_l - v_{i^*})/T}} \\ &= \frac{1}{1 + \sum_{l \neq i^*} e^{\text{negative number}/T}} \\ &\rightarrow \frac{1}{1 + \sum_{l \neq i^*} 0} \quad (\text{as } T \downarrow 0) \\ &= 1 \end{aligned}$$

In a LLM is it used as the last activation function of the neural network to normalize the output of a network to a probability distribution over all possible tokens in the vocabulary. It predicts the next word using this distribution. For example, the output of the neural network where the sentence to complete is:

The dog chased the ...

might score the next words like this:

“ball” $\rightarrow$ 4.0

“cat” $\rightarrow$ 2.0

“banana” $\rightarrow$ -1.0

Using the softmax function with say $T = 1$ will turn that into probabilities as:

“ball” $\rightarrow$ 0.875

“cat” $\rightarrow$ 0.118

“banana” $\rightarrow$ 0.007

Then the system might pick the highest-probability token or samples from the distribution (depending on settings).

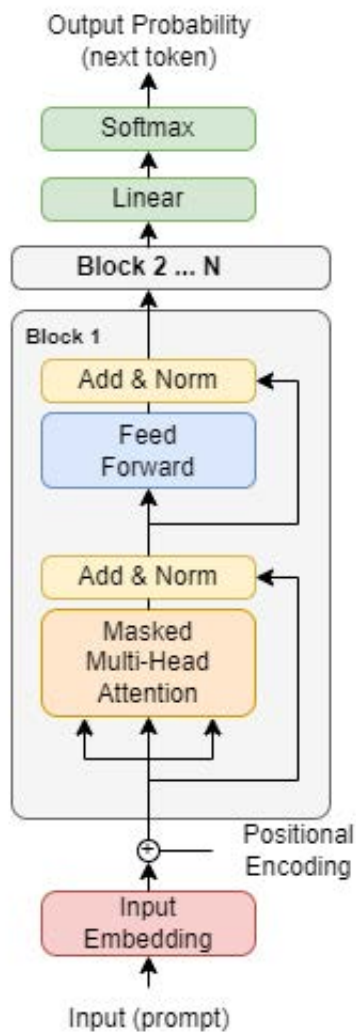


Figure 0.2: Architecture of the transformer model in LLMs